To Eliminate the Concern of Ride height Not ok / variation in Scorpio SCDC model

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To Eliminate the Concern of Ride height Not ok / variation in Scorpio SCDC model."

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	April'19	April'19
Completion Date	AUG'19	AUG'19

Current Status: **Closed**

Cross Functional Team

SNo	Description	Name	Designation	Dept	Email
1	Team Leader	Sayali More	Asst. Manager	Process	MORE.SAYALI@Mahindra.com
2	Member 1	Sumit Sisodia	Manager	Process	SISODIA.SUMIT@Mahindra.com
3	Member 2	Satish Pagar	Asst. Manager	MQA	PAGAR.SATISH@Mahindra.com
4	Member 3	Amit Tile	Deputy Manager	Mfg	TILE.AMIT@Mahindra.com
5	Member 4	Tushar Patil	Manager	Maint	PATIL.TUSHAR@Mahindra.com

Step1. Select Subject

Brief Description : To Eliminate the Concern of Ride height Not ok / variation in Scorpio SCDC model

What : To Eliminate the Concern of Ride height Not ok / variation in Scorpio SCDC model

Where : Field complaint reported

When : Since Feb-19

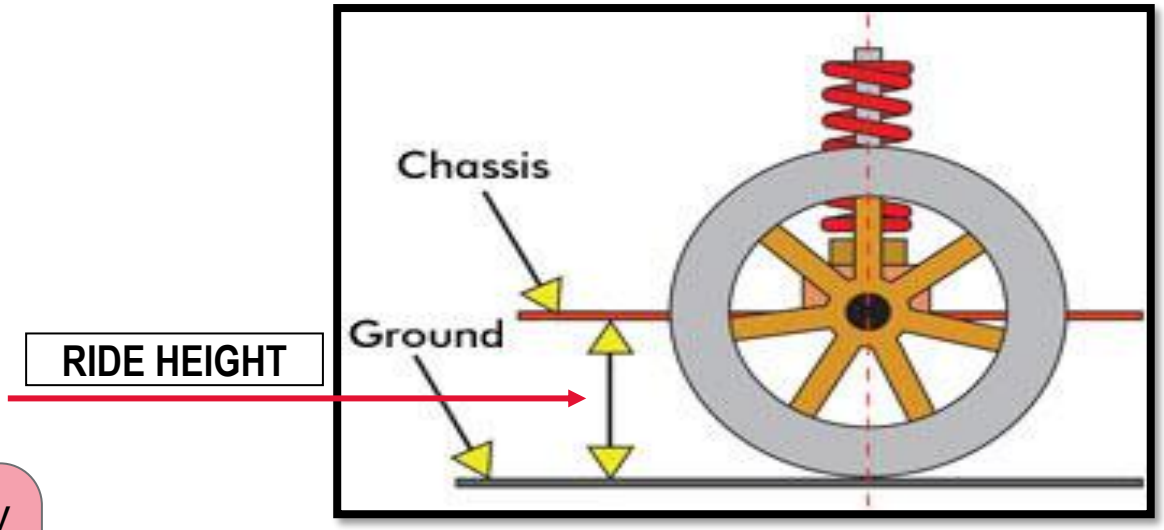
Why : It is a aesthetic factor which directly leads to dissatisfaction of customer

Who : ANOVAC check man, Quality inspector

How : 1 Incidence in field and 1 in RFI audit

Step1. Select Subject

Ride height (ground clearance) is the amount of space between the base of an automobile tire and the lowest point (typically the axle); Ground clearance is a critical factor in several important characteristics of a vehicle. Variations in clearance represent a trade-off between handling and vehicle's directional stability when moving in steady state condition. Vehicles directional stability is directly proportional to the safety factor.



Variable ride height directly affects the aesthetics & wheel alignment of the vehicle leading to customer dissatisfaction

2. Justify the choice (reasons)

PROMISE 2022
AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment
'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV
Doubling volume from overseas markets

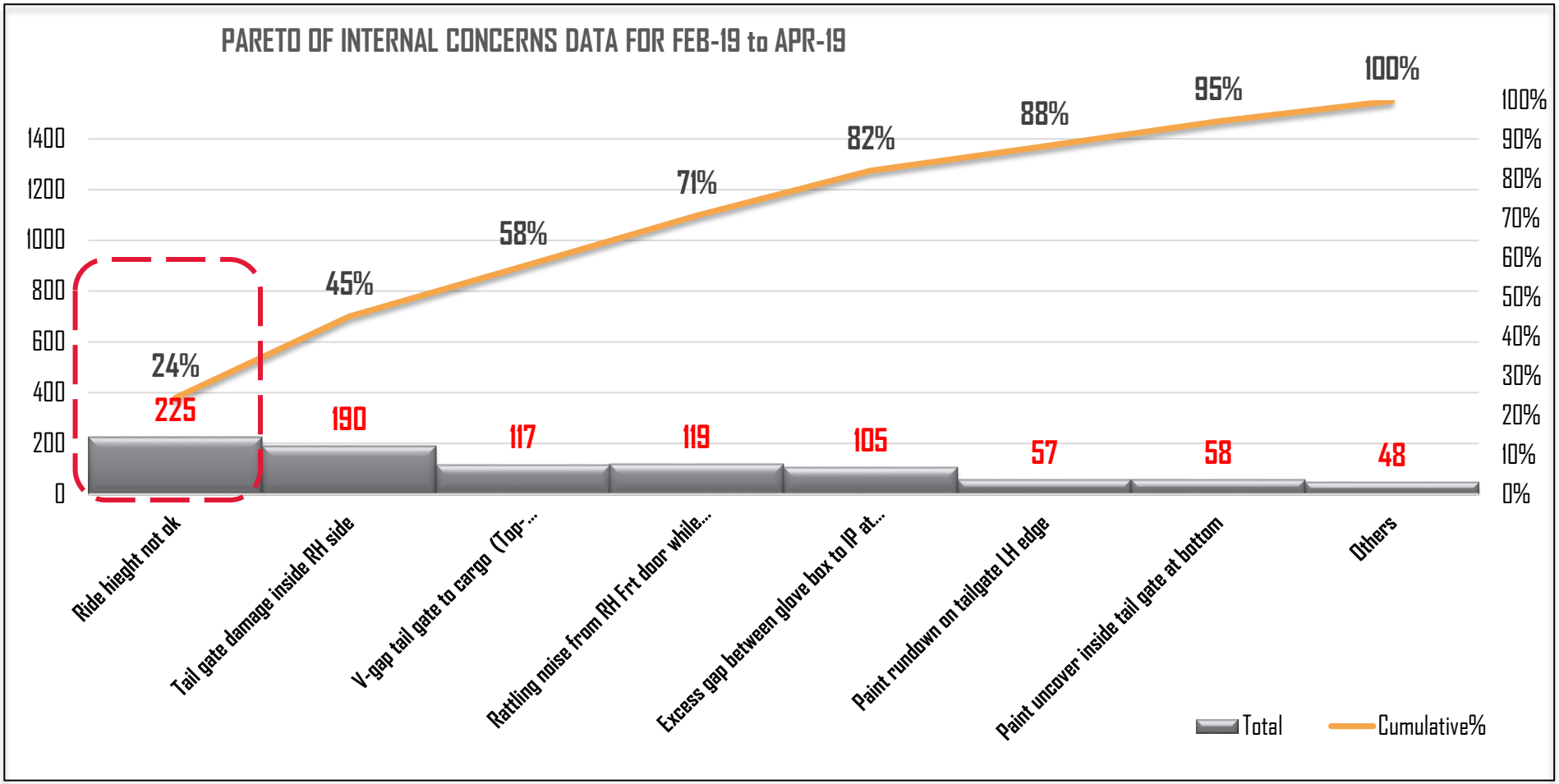
Leading in next generation mobility solutions

#InspiringPeopleToRise

Crusade 2.0

Departmental Objective :-
– DPR Improvement
- Offline rework reduction

2. Justify the choice



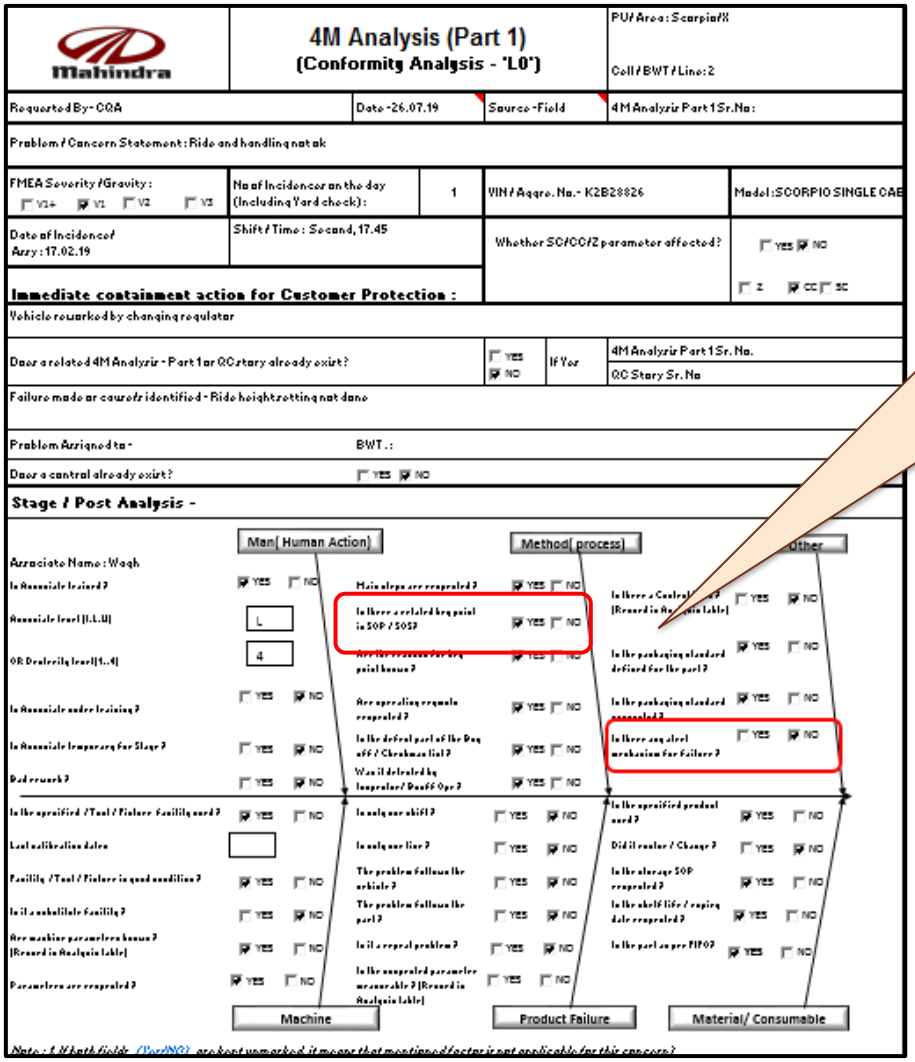
INFERENCE : Ride height not ok (variability) is top contributing defect leads to major STR loss at STCF.

2. Justify the choice

Severity & Consequences of Concern

- If vehicle with variable ride height issue goes in field then it will directly affect on the wheel alignment of the vehicle & may lead to accident.
- It will also affect the aesthetics of the vehicle.
- Due to such issues brand image of Mahindra will also gets violated.

Step 3. Understanding Current Situation


Mahindra

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/Area: Scorpio/K
Cell/BWT/Line: 2

Requested By - CQA | Date - 26.07.19 | Source - Field | 4M Analysis Part 1 Sr.No:

Problem / Concern Statement: Ride and handling not ok

FMEA Severity / Gravity: ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidence on the day (Including Yard check): 1 | VIN / Aggre. No. - K2B28826 | Model: SCORPIO SINGLE CAB

Date of Incidence / Arry: 17.02.19 | Shift / Time: Second, 17.45 | Whether SC/CC/Z parameter affected? ☐ Yes ☑ NO ☐ Z ☑ CC ☐ SC

Immediate containment action for Customer Protection :

Vehicle reworked by changing regulator

Does a related 4M Analysis - Part 1 or QC story already exist? ☐ Yes ☑ NO | If Yes: 4M Analysis Part 1 Sr. No. / QC Story Sr. No

Failure mode or cause is identified - Ride height setting not done

Problem Assigned to - | BWT.:

Does a control already exist? ☐ YES ☑ NO

Stage / Post Analysis -

Man (Human Action) | Method (process) | Other

Associate Name: Waqh

Is Associate trained? ☑ YES ☐ NO

Associate level (L.L.U) L

OR Dealership level (1..4) 4

Is Associate under training? ☐ YES ☑ NO

Is Associate temporary for Stage? ☐ YES ☑ NO

Bad remark? ☐ YES ☑ NO

Main steps are respected? ☑ YES ☐ NO

Is there a related key point in SOP / SOS? ☑ YES ☐ NO

Are the reasons for key point known? ☑ YES ☐ NO

Are operating requests respected? ☑ YES ☐ NO

Is the defect part of the Day off / Checksum list? ☑ YES ☐ NO

Was it detected by Inspector / Day off Opr? ☑ YES ☐ NO

Is there a Control plan (Record in Analysis table)? ☐ YES ☑ NO

Is the packaging standard defined for the part? ☑ YES ☐ NO

Is the packaging standard respected? ☑ YES ☐ NO

Is there any alert mechanism for failure? ☐ YES ☑ NO

Is the specified / Tool / Fixture facility used? ☑ YES ☐ NO

Last calibration date

Facility / Tool / Fixture in good condition? ☑ YES ☐ NO

Is it a suitable facility? ☐ YES ☑ NO

Are machine parameters known? (Record in Analysis table) ☑ YES ☐ NO

Parameters are respected? ☑ YES ☐ NO

Is only one shift? ☐ YES ☑ NO

Is only one line? ☐ YES ☑ NO

The problem follows the vehicle? ☐ YES ☑ NO

The problem follows the part? ☐ YES ☑ NO

Is it a repeat problem? ☐ YES ☑ NO

Is the unspecified parameter measurable? (Record in Analysis table) ☐ YES ☐ NO

Is the specified product used? ☑ YES ☐ NO

Did it evolve / Change? ☐ YES ☑ NO

Is the storage SOP respected? ☑ YES ☐ NO

Is the shelf life / expiry date respected? ☑ YES ☐ NO

Is the part as per PIP03? ☑ YES ☐ NO

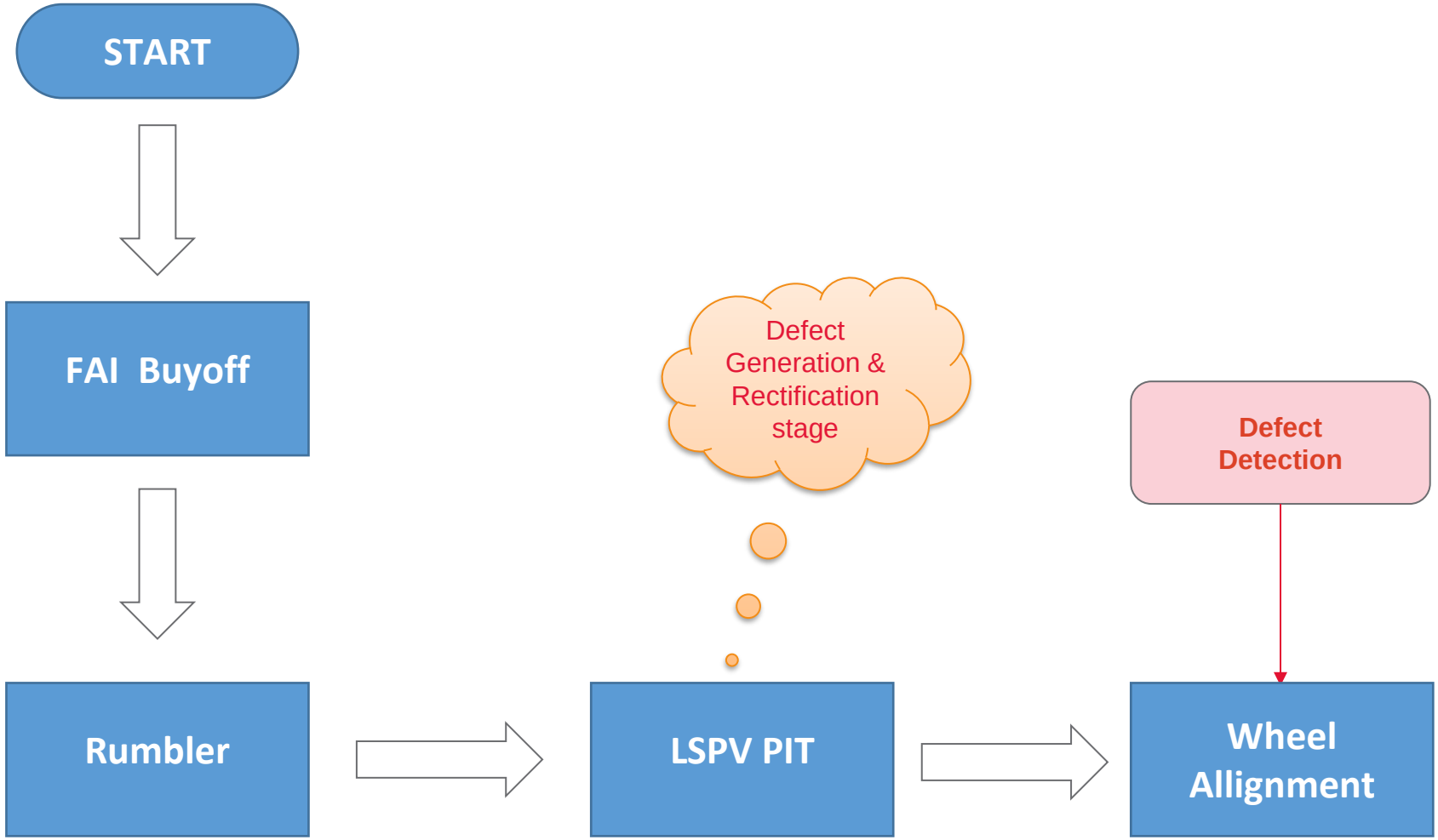
Machine | Product Failure | Material/ Consumable

Note : If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern.

Inference from L0,
Is no detection mechanism for failure & Associate did not set ride height of vehicle

4M analysis Good and Not so Good observations	
MAN	Associate skipped the vehicle without setting ride height
METHOD	No detection mechanism for failure
OTHERS	Control plan adhered on stage
MACHINE	Tools used are calibrated
PRODUCT	Problem follows the part
MATERIAL	handling of parts is no issue

Step 3. Understanding Current Situation



3. Understand the current situation

Setting of ride height

The Measuring gauge kept at both LH and RH LCA bolts on the flat surface provided.

Ride height setting is done by pulse tool. Simultaneously by tightening or loosening of the torsion bar bolt to get the required ride height.

Vehicle on stage (LSPV PIT)

1

2

The setting activity is carried out until the bolt setting qualifies the gauge. And when the bolt setting qualifies the gauge the associate on stage put a ride height OK stamp on the traveler card and the vehicle goes for further process indicating process complete.

3

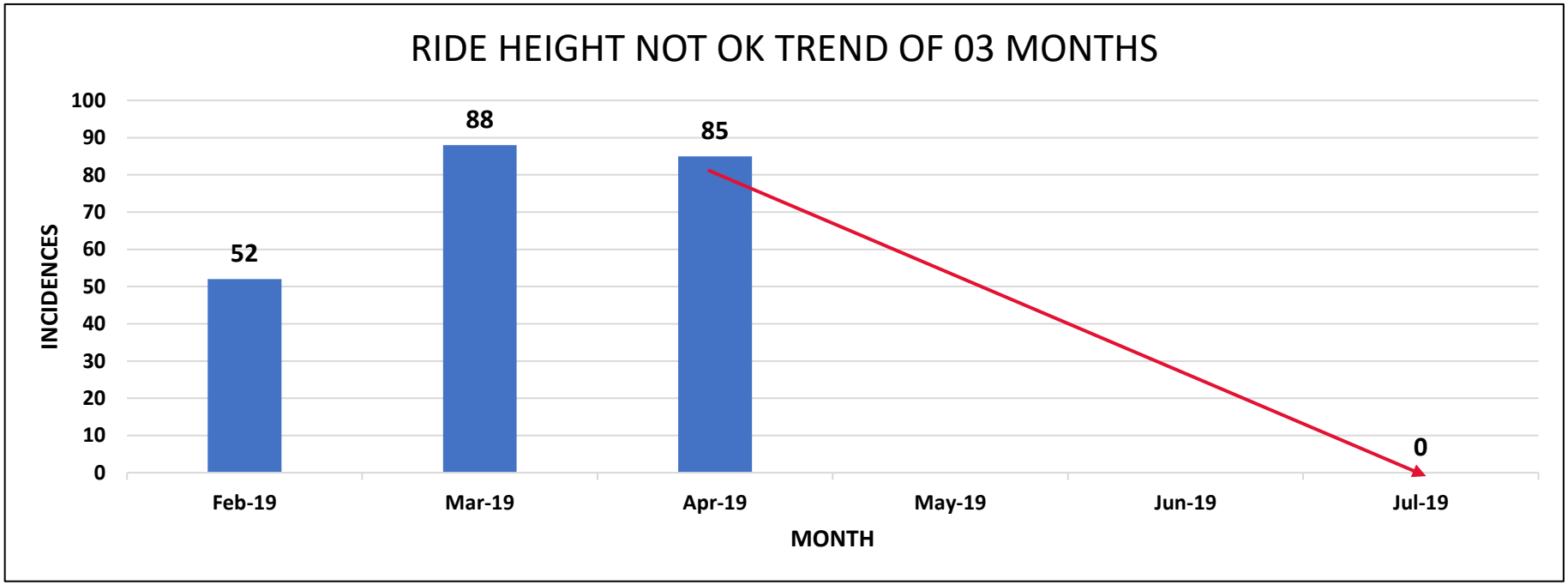
3. Understand the current situation

- Concern Reported in the field at dealer end, Ride and handling not ok
- **All L3/ L4 Operators Working** on Stage.
- Concern Reported in Scorpio SCDC model of multi model TCF.
- Concern reported as Ride and handling not ok.
- **Rework** is done at the ride height setting stage as re ride height setting - **not possible in tact time at stage**.
- **Rework** Takes more than **10 mins** time.

4. Select targets

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	**225 incidences reported in 03 months**
Target :	**Concern elimination by end of July - 19**



5. Analysis

5.1 Brain Storming

Brain Storming Team

Sayali More	Sumit sisodia
Satish Pagar	Amit tile
Tushar patil	Mandar joshi

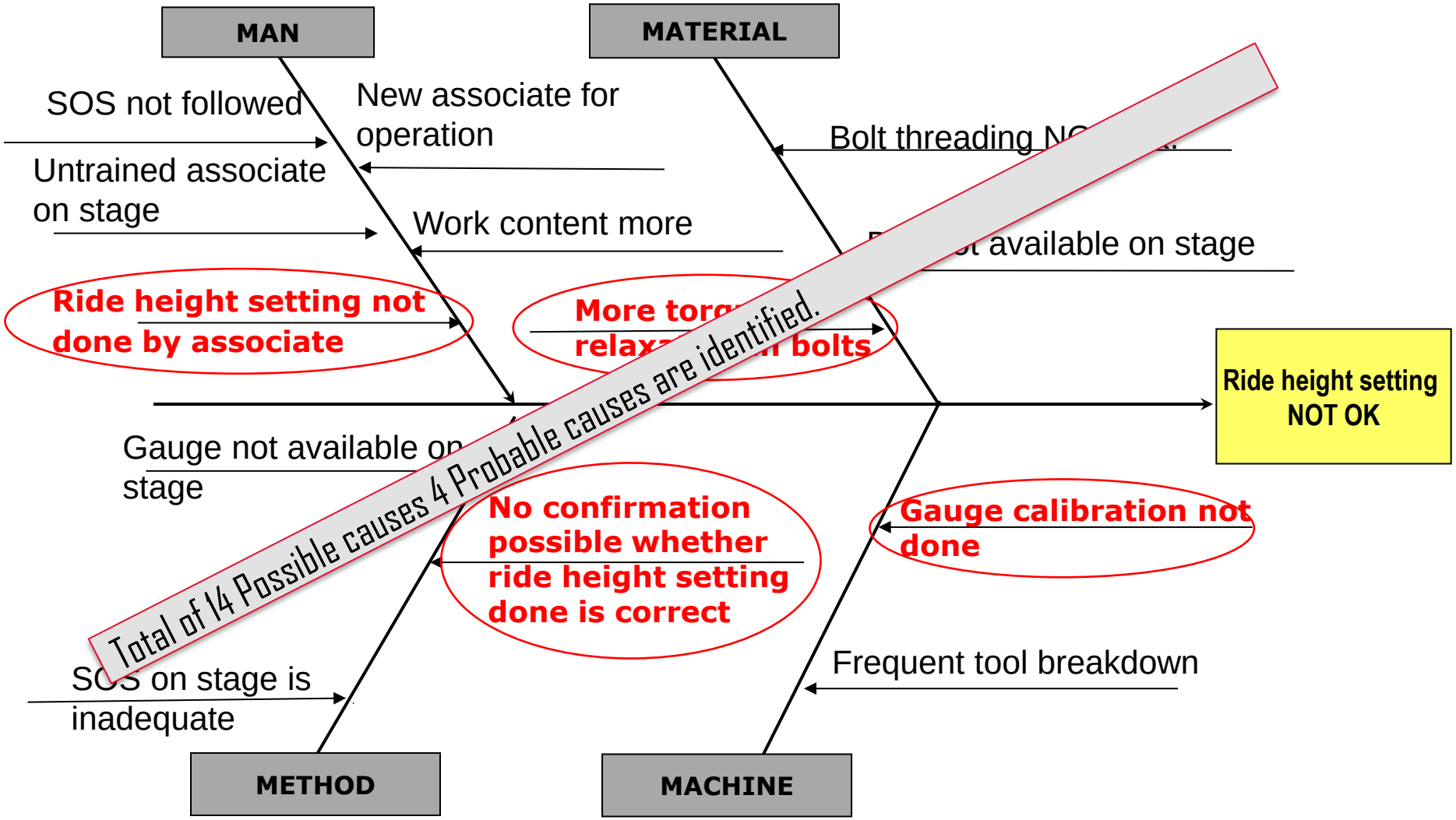
5. Analysis

List of Probable causes identification in Brain storming session

Sr. No.	Possible cause
1	SOS not followed.
2	Gauge not available on stage
3	Gauge calibration not ok
4	New Associate for height setting
5	Bolt threading not ok
6	Untrained Associate on stage.
7	Work content more.
8	More torque relaxation in bolts
9	SOS Inadequate
10	No certification possible whether setting done is correct.
11	Ride height not set by associate on stage
12	Insufficient torque delivered by the tool
13	Tool not available for tightening
14	Frequent tool breakdown

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Possible Causes	Sayali	Sumit	Sachin	Sor	Hari	Mandar	Nikhil	Total
1	SOS not followed.	1	3	1	1	3	1	1	11
2	Gauge not available on stage	1	3	1	3	3	1	1	14
3	Gauge calibration not ok	9	3	3	9	3	1	9	37
4	New Associate for height setting	3	1	1	1	3	1	1	11
5	Bolt threading not ok	0	3	3	2	2	0	0	10
6	Untrained Associate on stage.	3	3	3	3	3	3	3	21
7	Work content more.	1	1	3	3	3	3	3	17
8	More torque relaxation in bolts	3	3	3	3	9	9	9	39
9	SOS Inadequate	0	3	3	1	1	3	3	14
10	No confirmation possible whether setting done is correct.	9	9	3	3	9	9	9	51
11	Ride height not set by associate on stage	9	3	9	9	9	3	3	45
12	Insufficient torque delivered by the tool	0	3	2	2	2	3	2	14
13	Tool not available for tightening	0	3	3	2	2	3	3	16
14	Frequent tool breakdown	3	3	3	3	3	3	3	21

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 37.

Step 5. Analysis

List of Probable causes identification in Brain storming session

The possible causes who are having the Total score above 30 or more are considered as the probable causes. In Total of 14 Possible causes 4 Probable causes are identified.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr no	Not considered Causes	Reasons
1	SOS not followed.	SOS Followed by associate
2	Gauge not available on stage	Required gauge is available on stage
3	New Associate for height setting	All Trained associate
4	Bolt threading not ok	Bolt threading is OK
5	Untrained Associate on stage.	Trained associate
6	Work content more.	Work content is within the tact
7	SOS Inadequate	Checked found OK
9	Insufficient torque delivered by the tool	Found no issue regarding torque delivered
10	Tool not available for tightening	Required tool available on stage
11	Frequent tool breakdown	As checked no frequent breakdowns registered

5.1 Identify Probable cause

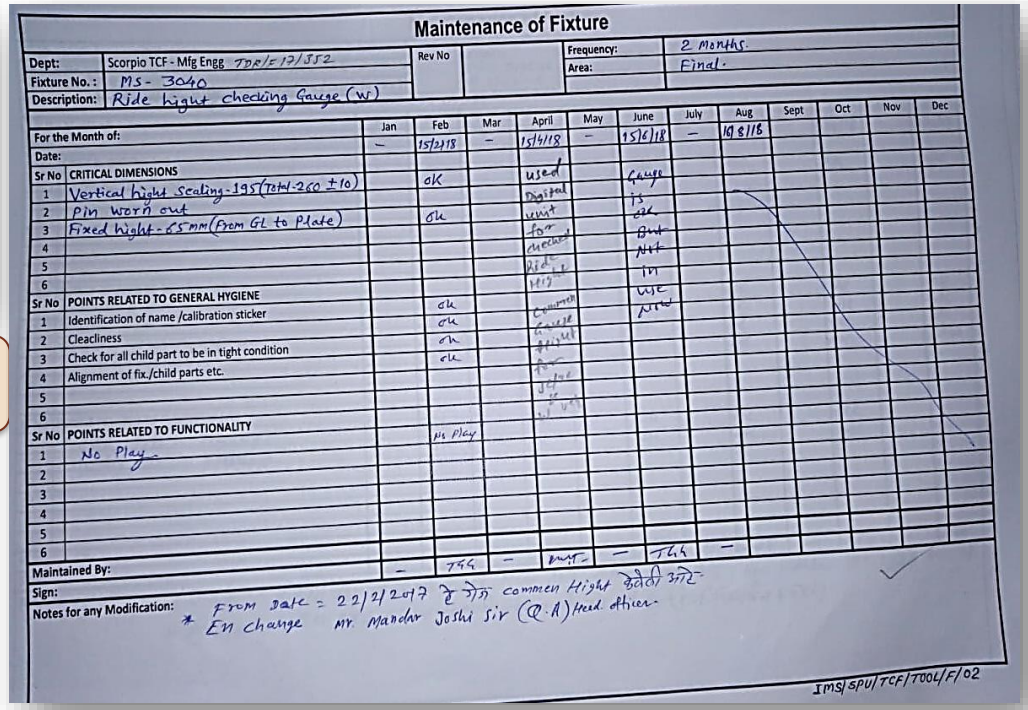
No	List of Probable Causes
1	Gauge calibration not done
2	Ride height setting not done by associate
3	No certification possible whether the set ride height is correct
4	More torque relaxation in bolt

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Gauge calibration not done	Checked the calibration record with tooling team	All records of calibration found OK as per plan	**Hypothesis Invalid**

"Setting gauge" calibration found OK



Maintenance of Fixture

Dept: Scorpio TCF - Mfg Engg TDR/F 17/552
Fixture No.: MS - 3040
Description: Ride hight checking Gauge (W)
Rev No
Frequency: 2 Months
Area: Final

For the Month of:	Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec
Date:	–	15/2/18	–	15/4/18	–	15/6/18	–	16/8/18				
Sr No CRITICAL DIMENSIONS												
1 Vertical hight Scaling-195(Total-260 ±10)		OK		used		Gauge						
2 Pin worn out				Digital		is						
3 Fixed hight-65mm(from GL to Plate)		OK		vernt		OK						
4				for		But						
5				checked		Not						
6				Ride		in						
				Hight		use						
Sr No POINTS RELATED TO GENERAL HYGIENE						Attd						
1 Identification of name /calibration sticker		OK		Common								
2 Cleacliness		OK		Gauge								
3 Check for all child part to be in tight condition		OK		Hight								
4 Alignment of fix./child parts etc.		OK		for								
5				Jeep								
6												
Sr No POINTS RELATED TO FUNCTIONALITY												
1 No Play		No Play										
2												
3												
4												
5												
6												
Maintained By:	–	JGG	–	[illegible]	–	JGG	–					
Sign:												

Notes for any Modification: * From Date = 22/2/2017 से जेग common Hight कैसे आये
En change Mr. Mandar Joshi Sir (Q.A) Head. Officer.

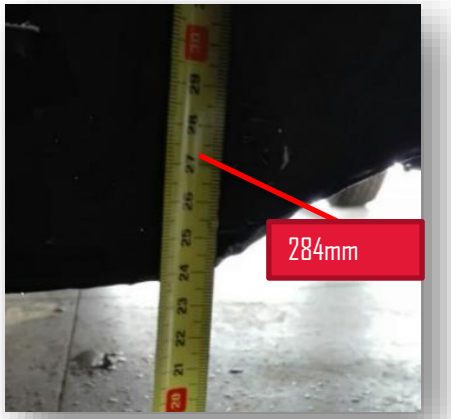
IMS/SPU/TCF/TOOL/F/02

Inference: All calibration records found OK as per the plan

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Ride height setting not done by associate.	Data analysis of 30 vehicles done	30 vehicles ride height measurement was done in CAI. Found 2 vehicles in which ride height was observed on higher side. On analysis found ride height not set on stage.	**Hypothesis valid**



Sr No	Ride Height			Sr No	Ride Height	
	LH	RH			LH	RH
1	268	267		16	269	268
2	263	263		17	270	266
3	268	267		18	265	264
4	266	256		19	268	268
5	268	269		20	268	267
6	266	270		21	263	263
7	264	265		22	268	267
8	268	268		23	266	256
9	284	281		24	267	268
10	265	266		25	278	279
11	268	267		26	268	269
12	263	263		27	270	268
13	268	267		28	265	268
14	266	256		29	269	268
15	263	263		30	264	265

30 Vehicles checked for ride height setting
2 cases are found not ok.
In both cases – The ride height was higher than the required limit

Inference: Ride height of 30 vehicles was measured in CAI, 2 vehicles found on higher side than required limit, on analysis found ride height was not set for said vehicles.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	No confirmation whether ride height set is correct or not	Process observation on gemba	On observation found, since there is no tool to certify the height setting, it is based on the associate present on the stage Subjective height setting done on gemba.	**Hypothesis valid**

Inference: No mechanism to certify the ride height set by associate on stage, Leading to variable ride height set by different associates.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	More torque relaxation in bolts	To check the torque of bolts on gemba using "Acta unit"	Vehicles checked for torque found OK. Torque found OK	**Hypothesis Invalid**

Bolt torque checking on gemba using "Acta Unit"

CHECKSHEET	
Description : Ride height setting	
Model : SCDC All	
Sr No.	Oservation/Result
1	OK
2	OK
3	OK
4	OK
5	OK
6	OK
7	OK
8	OK
9	OK
10	OK
11	OK
12	OK
13	OK
14	OK
15	OK
16	OK
17	OK
18	OK
19	OK
20	OK

Inference: Checked the torque of vehicles on gemba and found it to be OK as per specification of the tool.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr No	Probable Cause	Testing method	Conclusion
1	Ride height setting not done by associate.	30 vehicles ride height measurement was done in CAI. Found 2 vehicles in which ride height was observed on higher side. On analysis found ride height not set on stage.	**Hypothesis valid**
2	No confirmation whether ride height set is correct or not	On observation found, since there is no tool to certify the height setting, it is based on the associate present on the stage Subjective height setting done on gemba.	**Hypothesis valid**

2 Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Ride height setting not done by associate.	Associate forgot to set ride height.	No full proof mechanism for ride height setting	Manual ride height setting	Full proof mechanism to be provided for ride height setting
2	No confirmation whether ride height set is correct or not	Associate forget to confirm ride height	No full proof mechanism for ride height checking	Visual checking	Full proof mechanism for ride height checking

6. Implement the corrective measures

Possible Remedies Identified

Sr. No.	Root cause	Remedy Developed	Time to implement	Cost of Improve ment	Easily reversible	Complexity	Decision
1	Manual ride height setting	Full proof mechanism to be provided for ride height setting	Medium	Medium	yes	Low	**Accepted**
2	Visual checking	Full proof mechanism for ride height checking	Medium	Medium	yes	Low	**Accepted**

6. Implement the corrective measures

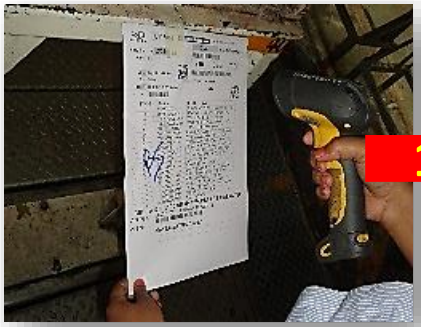
No	Root Cause	Action plan	Actual date	Responsibility	Status
1.	No alert mechanism for failure at stage	Alert mechanism provide at re-torqueing stage – Auto ride height setting system implemented on stage.	May-19	Sayali More	CLOSED

Details available in next slide.....

6. Implement the corrective measures

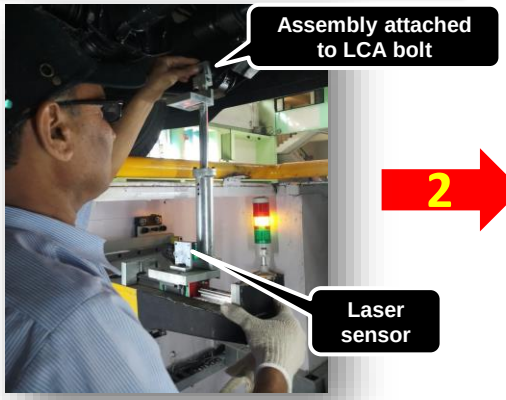
Alert mechanism – Auto ride height setting system implemented on stage

Scan the VIN no on the vehicle



1

Associate has to engage the laser sensors on the front LCA bolts



2

Associate has to then engage the servo motor assembly to the torsion bar bolt where ride height setting is done

when the cycle is completed, associate has to disengage the motor and press the work complete button (When cycle is completed the HMI screen provided shows the set reading and cycle complete message on screen, along with green indication on tower lamp)

3

Associate has to press the start push button and the cycle starts.(Here on when cycle starts the ride height is adjusted automatically by the motor assembly by moving in clockwise and anti-clockwise as required to set the desired height)

4

6. Implement the corrective measures

No	Root Cause	Action plan	Actual date	Responsibility	Status
2	No certification possible for Ride height correctly set.	To provide detection mechanism – Ride height OK sticker started for traceability and certification after process complete.	May-19	Sayali More	CLOSED

Certification and traceability provided at ride height setting stage

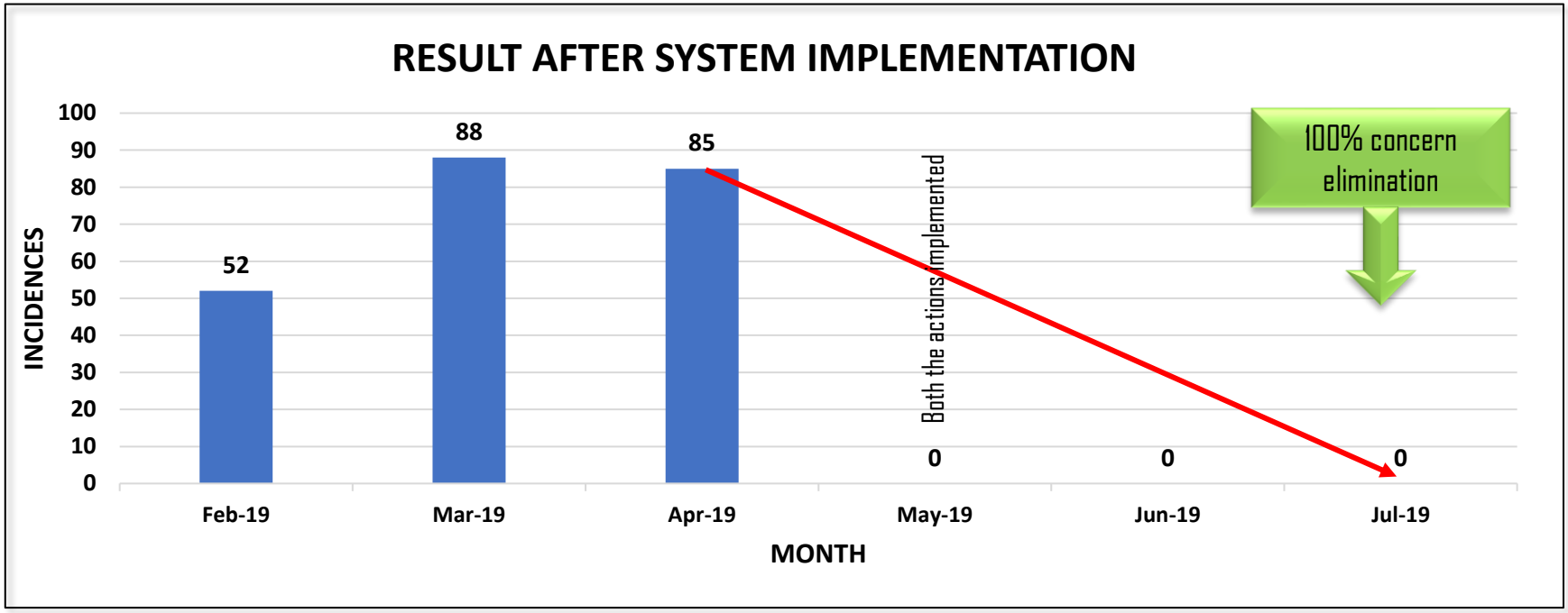
Complete the setting process

1. After process complete, the entire assembly has to be at home position
2.Ride height OK print received. Paste the sticker on the traveler card.
3. Data is saved in the controller against said VIN

2

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend



Inference – After Implementing auto ride height setting process for ride and handling NOT OK concern and traceability for the same monitored for 3 months No issue reported at Internal Audit and well as Field.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	**Actual date**	Responsibility	Status
1	Alert mechanism provide to be provided at ride height setting stage	28.04.19	Sayali More	CLOSED
2	Detection mechanism to be provided at setting stage	28.04.19	Sayali More	CLOSED
3	SOS Modification	25.04.19	Amit Tile	CLOSED
4	PFMEA updation	25.04.19	Amit Tile	CLOSED
5	Control Plan Updation	25.04.19	Vikrant Ghule	CLOSED

Check & confirm the results & effects

7.3 Tangible & Intangible benefits:

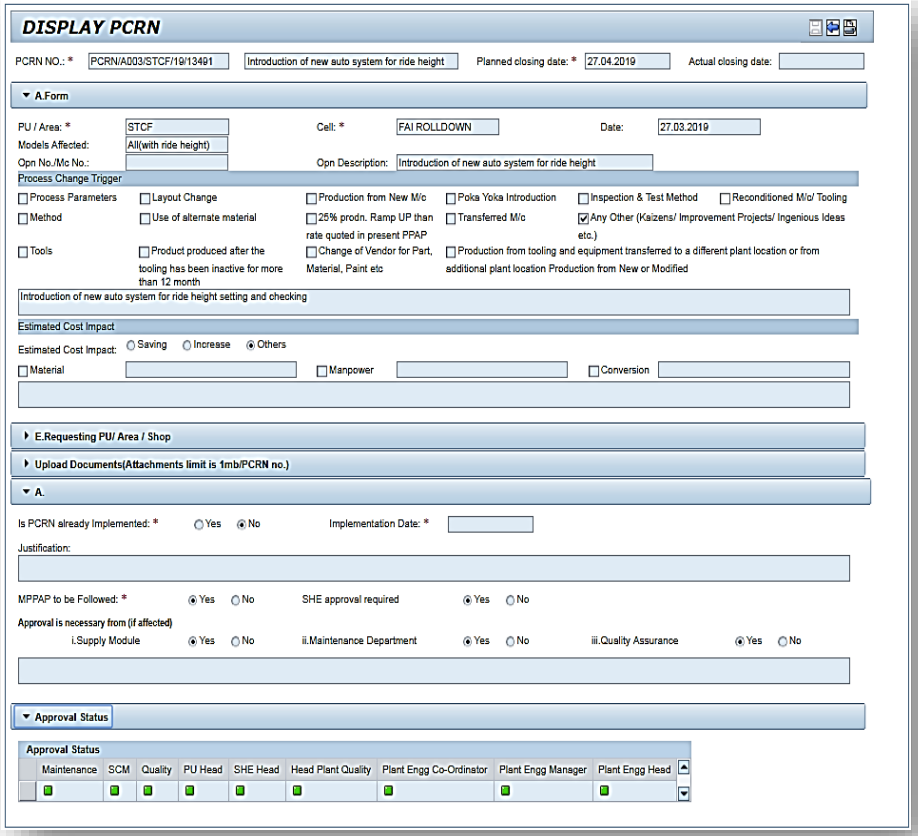
Tangible Benefit:-

- Subjective ride height checking to objective leading to real time data monitoring
- Field complaint eliminated.
- Traceability of 100% vehicles.
- STR loss due to Ride height not ok is eliminated.

Intangible Benefit:-

- Noise level of 102 decibels reduced to 10 decibels
- Associate Moral Improved.
- Associate fatigue reduced
- Deskilling of the associate on stage

Process Changed Request Note – Approved by all stockholders.



DISPLAY PCRN

PCRN NO.: * PCRN/A003/STCF/19/13491 | Introduction of new auto system for ride height | Planned closing date: * 27.04.2019 | Actual closing date:

A.Form

PU / Area: * STCF | Cell: * FAI ROLLDOWN | Date: 27.03.2019

Models Affected: All(with ride height)

Opn No./Mc No.: | Opn Description: Introduction of new auto system for ride height

Process Change Trigger

- [ ] Process Parameters
- [ ] Layout Change
- [ ] Production from New M/c
- [ ] Poka Yoka Introduction
- [ ] Inspection & Test Method
- [ ] Reconditioned M/c/ Tooling
- [ ] Method
- [ ] Use of alternate material
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Transferred M/c
- [x] Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- [ ] Tools
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Change of Vendor for Part, Material, Paint etc
- [ ] Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified

Introduction of new auto system for ride height setting and checking

Estimated Cost Impact

Estimated Cost Impact: ○ Saving ○ Increase ◉ Others

- [ ] Material
- [ ] Manpower
- [ ] Conversion

E.Requesting PU/ Area / Shop

Upload Documents(Attachments limit is 1mb/PCRN no.)

A.

Is PCRN already Implemented: * ○ Yes ◉ No | Implementation Date: *

Justification:

MPPAP to be Followed: * ◉ Yes ○ No | SHE approval required ◉ Yes ○ No

Approval is necessary from (if affected)

i.Supply Module ◉ Yes ○ No | ii.Maintenance Department ◉ Yes ○ No | iii.Quality Assurance ◉ Yes ○ No

Approval Status

Approval Status

Maintenance	SCM	Quality	PU Head	SHE Head	Head Plant Quality	Plant Engg Co-Ordinator	Plant Engg Manager	Plant Engg Head
■	■	■	■	■	■	■	■	■

8. Standardization

8.1 Updation of documents: e.g. DFMEA,PFMEA,DCP,SOP,SOS,drawing.

SN	Document	Actual date	Responsibility	Status
1	PCRN Implemented PCRN/A003/STCF/F19/13491	27.04.19	Sayali More	Closed
2.	SOS Modification.	25.04.19	Amit Tile	Closed
3.	PFMEA updation.	25.04.19	Amit Tile	Closed
4	Control Plan Updation	25.04.19	Vikrant Ghule	Closed

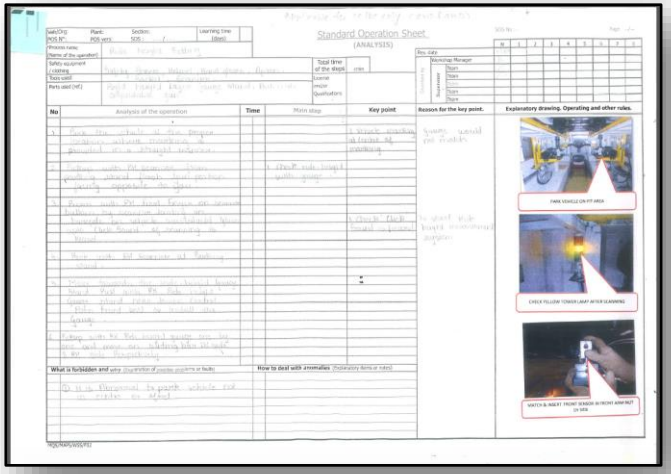
8.2 Training:

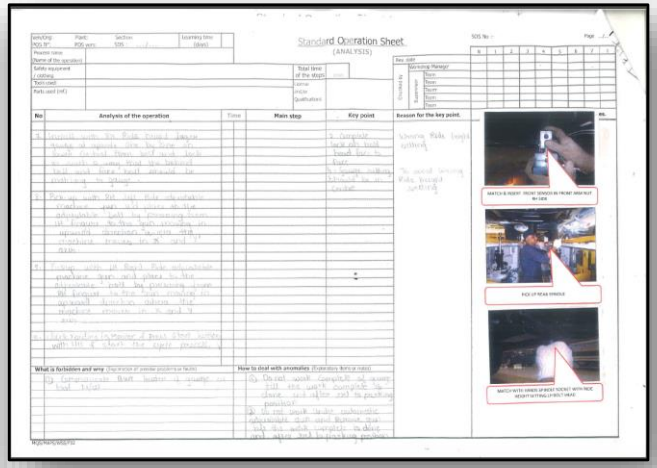
SN	Training Identified	Actual date	Responsibility	Status
1.	On job Training given to operator.	25.04.19	Amit Tile	Closed

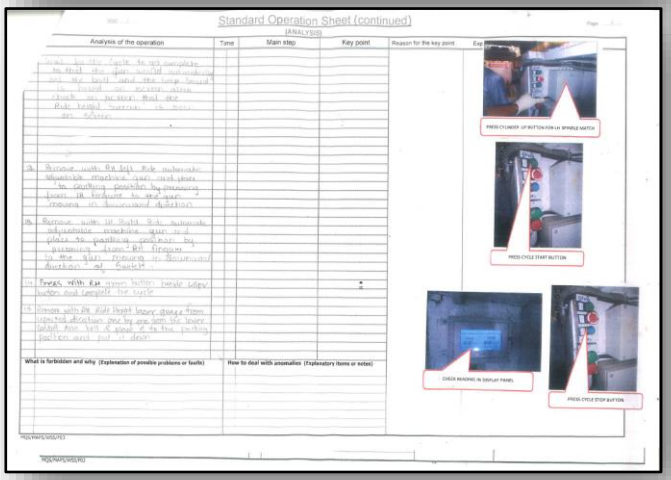
8. Standardization

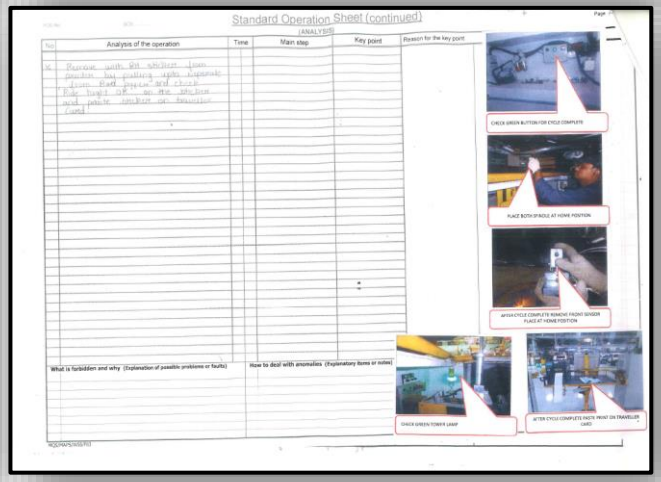
8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SOS Updation









SOS updated for Auto ride height setting for SCDC variants

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

PCRN implemented PCRN/A003/STCF/F19/13491

DISPLAY PCRN

PCRN NO.: * PCRN/A003/STCF/19/13491 | Introduction of new auto system for ride height | Planned closing date: * 27.04.2019 | Actual closing date:

A.Form

PU / Area: * STCF | Cell: * FAI ROLLDOWN | Date: 27.03.2019

Models Affected: All(with ride height)

Opn No./Mc No.: | Opn Description: Introduction of new auto system for ride height

Process Change Trigger

- [ ] Process Parameters
- [ ] Layout Change
- [ ] Production from New M/c
- [ ] Poka Yoka Introduction
- [ ] Inspection & Test Method
- [ ] Reconditioned M/c/ Tooling
- [ ] Method
- [ ] Use of alternate material
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Transferred M/c
- [x] Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- [ ] Tools
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Change of Vendor for Part, Material, Paint etc
- [ ] Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified

Introduction of new auto system for ride height setting and checking

Estimated Cost Impact

Estimated Cost Impact: ○ Saving ○ Increase ◉ Others

- [ ] Material
- [ ] Manpower
- [ ] Conversion

E.Requesting PU/ Area / Shop

Upload Documents(Attachments limit is 1mb/PCRN no.)

A.

Is PCRN already Implemented: * ○ Yes ◉ No | Implementation Date: *

Justification:

MPPAP to be Followed: * ◉ Yes ○ No | SHE approval required ◉ Yes ○ No

Approval is necessary from (if affected)

i.Supply Module ◉ Yes ○ No | ii.Maintenance Department ◉ Yes ○ No | iii.Quality Assurance ◉ Yes ○ No

Approval Status

Approval Status

Maintenance	SCM	Quality	PU Head	SHE Head	Head Plant Quality	Plant Engg Co-Ordinator	Plant Engg Manager	Plant Engg Head

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

FMEA Updation

Plant : Nashik			Model :Scorpio				*Mahindra & Mahindra Ltd.*								Cross Functional Team			
Part No/s.:			Part Name/ Process:												Suhas Aher	Bbab Awhale		
Latest Change Letter:			Optn No.:5150				*Process Potenial Failure Mode & Effect Analysis.*								Vikrant Ghule	Santosh Jadhav		
PU Shop :Scorpio TCF			Optn Desptn: Ride height setting				*Reviewed by :Suhas Aher*			Rev. No :	27	Doc No. PFMEA/SPU/SCORPIO/FINAL/5150			Sayali More			
Module/ Cell :Final			M/C Name & No.:				*Approved by : Mandar Joshi*			Rev. Date :	11.04.2019	FMEA Date (Orig) :20/9/00						
Process Function	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results				
														Action Taken	Sev	Occ	Det	RPN
Ride height setting.	Setting the ride height of the vehicle as per the model.(WSC/DC) -	Ride height setting wrong.	**Mfg. & Assy:** Camber value setting will be wrong , rework. (5) Veh dragging one side. **End user:** tyre wear. (7)	6		Improper power tool calibration	Periodic Tool Calibration 1) Fixed manpower for ride height setting. 2)Self certification of operator on traveller card by writing Ride height ok. 3) All camber rework to be recheck for ride ht.	4	Digital display for ride hight measurement	5	120	Introduction of new auto system for ride height setting	Sayali more 11.04.2019	Action completed				
4WD socket connection	Under body socket connection	Connection poor fit.	Mfg. & Assy: Rework. (5) End user: 4WD function will be malfunction. (7)	6		Operator error.	1) Fixed manpower for ride height setting and socket conn.. 2)Self certification of operator on traveller card bysigning on TC 3) Relative marking started at socket .	4	1) Fixed manpower for ride height setting and socket conn.. 2)Self certification of operator on traveller card bysigning on TC 3) Relative marking started at socket	5	120							

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

Control plan Updation

Plant : Nashik		Model :Scorpio& bM1		Mahindra & Mahindra Ltd.							Cross Functional Team		
Part No/s.: SNC26MBC8R010		Part Name/ Process: FAI Line		Control Plan							Vikrant Ghule	Pravin Patil	
Latest Change Letter:		Doc No.:		Prototype :		Prelaunch:		Productio			Suhas Aher	Taj Shaikh	
Aggregate No.:		Aggregate Part Descr		Module Name : Scorpio Assy		Reviewed by : Vikrant Ghule		Rev No: 153		Critical Char: <CC>	Vikrant Morade	Mandar Joshi	
PU Shop :Scorpio		Next Aggregate No.:		Cell Name : Final line		Approved by : Mandar Joshi		Rev. date: 25.12.17		Significant Char : <SC>		CP Date(Orig) : 15.12.2005	
Opn No	Operation Description	Device, Jig, Tools	No.	Product	Process	Ch	Product/Process specifications	Evaluation Measurement Tech.	Sample Size	Sample Freq.	Control Method Record	Control Method Resp.	Reaction Plan
52 0	Ride Height Setting - Reading (4WD Only)	separate Setting gauge for 'W' and 'SC/DC'	1	Ride hide setting	Pasting Ok stiker card		For W=264+/-8mm For SC/DC=264+/-8mm	Built-in for 2WD As per digital display and printout	100%	daily	Traveller card print out on traveller card	Associate TCF	Recheck & certify
			2		Engagment of sensor on ride height bolt		As per SOS	Visual	100%	daily	Traveller card print out on traveller card	Associate TCF	Recheck & certify
			3		Engaging of servo motor to ride height bolt		As per SOS	Visual	100%	daily	Traveller card print out on traveller card	Associate TCF	Recheck & certify
			4		Pressing start bolt		As per SOS	Visual	100%	daily	Traveller card print out on traveller card	Associate TCF	Recheck & certify
			5		Machine PM		As per checksheet	As per checksheet	100%	Once in 2 months	As per checksheet	Associate - PE	Re calibrate, confirm and record the er

8. Standardization

8.2 Training:

OPL - One Point Lesson — Mahindra

Theme	Auto Ride Height सेटींग	No	
		Date	16/8/19

Classification	Basic knowledge	Improvement cases	Trouble cases	Module Manager	Section Head	Cell Leader	Prepared By
	☑	☐	☐				

1) vehicle LSPV फीक्चर आल्यानंतर traveller card scan करणे.
2) पीटमधील LH व RH साईड चे sensor gauge Arm बोल्ट ला अडकवणे.
3) त्यानंतर LH व RH साईड चे Ride Height Adjustable spindle upward direction ला घेणे व spindle socket LH व RH Resp. बोल्ट हेड मध्ये अडकवणे.
4) त्यानंतर Display मध्ये Readings चेक करून start बटन दाबणे व cycle सुरू करणे.
5) cycle complete झाल्यावर Beep sound असा आवाज येऊ येईल त्यानंतर, दोन्ही LH व RH साईड चे Adjustable spindle खाली करून Home position ला ठेवणे.
6) त्यानंतर LSPV Button च्या बाजुला असलेले Green Button दाबणे.
7) त्यानंतर LH व RH साईड चे sensor gauge खाली काढून Home position ला ठेवणे.
8) त्यानंतर OK print traveller card ला लावून vehicle पुढील स्टेज वर पाठवणे.

Actual Results										
Date		SGB	PDN	SSN	AKA	ADS	BDB	SKM	RGN	RBN
Trainer										
Trainee		BBS	DDM	AAJ	SBN					

Operator communication and On job Training given to operators : Resp Amit tile

Operator communicated for the new process of ride height setting and checking

9. Summaries & Plan future actions

9.1 Conclusion:

Elimination of the ride and handling not ok Concern by Taking action on all the Probable causes of Ride height setting not done by associate and No confirmation whether ride height set is correct or not

9.2 Horizontal Deployment:

System given for implementation at Nashik Bolero Line, Haridwar, Zahirabad & Chakan plants

9.3 Lessons Learnt:

Things Gone Wrong : Associate complexity not considered while implementing manual setting and checking system for ride height.

Things Gone Right : Use of digitization application help us to maintain & monitore the results

Thank you

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